

Oncogenic Viruses - Factoid Sheet

Week 8 • Core Lecture 5 • current Thai deck controls

Master map: oncogenic viruses either press the gas (growth signaling/oncogene activation) or cut the brakes (p53/RB/apoptosis control).

General Principles

Concept	Lecture meaning
Proto-oncogene	Normal growth-promoting gene; can become oncogenic when activated/dysregulated
Oncogene	Activated growth signal driving proliferation
p53	DNA-damage repair/apoptosis checkpoint
RB	Cell-cycle brake; loss permits inappropriate progression
Provirus	Viral genome integrated into host chromosome

EBV

LMP-1 → CD40-like growth signal + **BCL2** survival • **EBNA-2** → cyclin D/host-gene activation • **vIL-10** → reduced T-cell activation.

EBV association	Key cue
Burkitt lymphoma	EBV sets proliferative stage; MYC t(8;14) ultimately drives cancer
Hodgkin / NHL	Within EBV neoplasia spectrum
Nasopharyngeal carcinoma	Geographic association emphasized
Oral hairy leukoplakia	EBV-associated condition in AIDS
Mono	Atypical/reactive CD8 T cells; heterophile antibodies

Deck nuance: slide 14 says EBV is **not directly oncogenic in Burkitt**; it acts as a polyclonal B-cell mitogen that helps set up MYC translocation and additional mutations.

Handout-only board link: EBV uses CD21 on B cells; endemic Burkitt may show jaw mass + starry-sky histology.

HPV

Type	Disease
1-4	Skin / plantar warts
6, 11	Condylomata acuminata
16, 18, 31, 33	High-risk cancer types in deck

E6 → p53 loss • **E7** → RB loss • **hallmark histology** = koilocytosis.

HHV-8 + Other Oncogenic Viruses

HHV-8 / KSHV

Viral factor	Current-deck effect
Viral BCL2	Anti-apoptosis
LANA-1 / cyclin	Interferes with RB-linked control
vFLIP	Inhibits apoptotic signaling
Viral IL-6 + IL-6/IL-8/chemokines	Growth signaling for B cells and endothelial spindle cells
vIRF	Prevents IFN repression of c-MYC

Kaposi = violaceous plaques + proliferating endothelial spindle cells + slit-like vascular spaces containing RBCs.

HBV / HCV / Retroviral Logic

Virus	Lecture mechanism
HBV	Repeated injury/repair → fibrosis → HCC; genome integration may activate oncogenes, destabilize chromosomes, disrupt cell-cycle regulators
HCV	Listed as oncogenic RNA virus / HCC association
Retrovirus	Proviral insertion can activate adjacent proto-oncogenes; slide discusses deletion/transduction/insertional mechanisms

Handout-only board link: HTLV-1 → adult T-cell leukemia/lymphoma via Tax. The current slide 7 uses a different HTLV disease list, so keep the two sources distinct.

Treatment / Prevention

Virus	Current-deck approach
HPV warts	Imiquimod, podofilox, sinecatechins, cryotherapy, surgery
HPV prevention	Pap screening + Gardasil-9 (6,11,16,18,31,33,45,52,58)
Kaposi	Radiation/chemotherapy; ART greatly reduced HIV-associated KS; slide also lists interferon-α and additional therapy
EBV	Deck lists symptomatic + acyclovir; no vaccine. Current Herpesvirus lecture conflicts on EBV antiviral benefit.

Vignette Shortcuts

- African jaw tumor + MYC 8;14 → **EBV/Burkitt**
- Koilocyte / cervical dysplasia → **HPV**
- Cauliflower genital wart → **HPV 6/11**
- Violaceous lesion + spindle cells/slit spaces → **HHV-8/Kaposi**
- Chronic viral hepatitis + HCC → chronic injury/regeneration; HBV can also integrate

Final anchors: EBV = LMP-1/EBNA-2 + MYC • HPV = E6/p53 + E7/RB + koilocyte • HHV-8 = anti-apoptosis/growth signals + Kaposi spindle vessels.

Primary: current 36-slide Thai deck. MCG 2029 AI Notes “Lecture 9: Oncogenic Viruses” is labeled secondary enrichment.